

# Determining Business Process Improvement Priorities at Surabaya City Office for Population Administration & Civil Registration

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**Abstract:** Organizations often face difficulty in the evaluation and prioritization of their business processes. Many performance measurement indicators are defined at the aggregate and not at the process level. Surabaya City Office for Population Administration & Civil Registration (COPACR) is a local government agency which face these challenges. This paper attempts to solve COPACR challenges by applying BPM approach. First, process performance measurement guideline is developed. Next, the business process selection stage is carried out. The last stage is composing performance measurement indicators. From the business process selection stage, the priorities for process improvement initiatives are application processes for Birth Certificate, Biodata Change on Family Cards with the KLAMPID Application, and (3) Indonesian Citizens Transfer Certificate of Inter-City/Regency/Province processes certificate. In addition, the development of performance measurement indicators results in 29 performance measurement indicators related to Birth Certificate business processes.

**Keywords—***Business Process Management, Process Selection, Process Performance Measure, Process Portfolio, Performance Measurement Indicators.*

## I. INTRODUCTION

Organizations today must find ways to continuously improve their business processes to achieve their goals. Business process is a set of activities, events, processes, and decision points carried out by a company or organization by involving one to several actors and/or objects in generating certain values or benefits for customers. Business process management (BPM) is a set of methods, techniques, and tools in identifying, discovering, analyzing, redesigning, implementing, managing, monitoring, and evaluating business processes to maximize the performance of the company or organization as a whole. The interrelated phases of process identification, discovery, analysis, redesign, implementation, and monitoring carried out on an ongoing basis is referred to as the BPM Lifecycle [1].

Many organizations face different challenges in different BPM lifecycle. One of the challenges that organizations face is the evaluation of their business processes. This aspect is important because it helps the company or organization in determining and measuring progress to achieve the goals that have been set and ensuring that the BPM initiatives contribute to the organizations value. Performance

measurement indicators enable organizations or agencies to identify and prioritize areas of improvement and use of resources more efficiently [2]. Performance measurement indicators can provide a "clear signal" about the progress, success, and failure of a defined policy and program [3].

Information related to performance also provides a clear signal for the company or organization to learn more from the performance results and develop and implement strategies that can be done to improve that performance [2]. Therefore, it is very important to collect and ensure that performance information is updated regularly so that it can provide an accurate picture of the performance of the organization [4]. The frequency of updating performance information in general will in most cases depend on the services and programs provided by the company or organization [2].

According to [5], there are many frameworks used in performance measurement such as the Balanced Scorecard (BSC), Key Performance Indicators (KPI), and others. However, this research article only discusses performance measurement indicators in general and not associated with business processes. This fact raises several issues. First, the absence of process-level performance measure means that the priority for improving business processes is carried out on an ad-hoc basis.

Organizations in the public sector has unique characteristics that differentiate them from private companies. The wave of digital transformation, accelerated by the COVID-19 pandemic have forced the public sector organizations to innovate their product and services in serving the public. However, such organizations are also constrained in terms of structure and human resource capacity. Public sector organizations must prioritize their improvement efforts to ensure that they can achieve the intended goals.

Surabaya City Office for Population Administration & Civil Registration (COPACR) is a local government agency that provides population administration and civil registration services [6]. Presidential Regulation of the Republic of Indonesia Number 95 of 2018 encourages local government agencies, especially Surabaya COPACR to apply a business process-oriented approach on daily operational activities [7]. COPACR have launched various initiatives to improve their

services to the public. However, they still received many complaints. COPACR top management faces difficulty in choosing and prioritizing the improvement initiatives that they must carry out in the future. COPACR also does not have clear criteria in measuring the performance of actors and business processes.

In this paper, we attempt to apply BPM approach in the case of COPACR. First, we develop process performance measurement (PPM) guideline so COPACR and other public organizations can measure their performance at the business process level. Second, we use the PPM to develop Process Portfolio that the organizations can use to prioritize their improvement initiatives. Results of this paper contribute to the PPM literature by providing a guideline for PPM development in the public sector. The paper contributes practically by applying PPM as a basis to prioritize business process improvement in COPACR, and similar public sector organizations in Indonesia.

This paper is organized as follows. First background on prioritization of business process improvement and process performance measurement is provided in section 2. This is followed by the methodology including the development of PPM guidelines and its implementation in COPACRS case. Then section 4 describes the results and discussion. Finally, the conclusion is given in the last section.

## II. BACKGROUND

### A. Previous Study

Two studies have been done that relate to business process improvement in public sector in Indonesia. In [8], the authors discuss about practice to implement business process map in 10 government institutions in Indonesia. However, this study does not discuss about how to conduct business process selection. In [9], the authors design a performance measurement system at the Southeast Asian Ministers of Education Organization Regional Center for Food and Nutrition - SEAMEO RECFON using four perspectives on the balanced scorecard. However, this study does not focus specifically on business process performance.

### B. Prioritizing Business Process Improvement

a) *Process Selection*: Process Selection is a step in process discovery phase of BPM lifecycle that assess the performance of organization business process. In the process selection, there are several aspects of the criteria used, including aspects of strategic importance, health, and feasibility [1].

b) *Process Portfolio*: Process Portfolio is a visualization of business processes performance based on their strategic importance and health condition. The feasibility to improve the business processes are shown in different colors of the dots [1].

### C. Process Performance Measurement

Process Performance Measure is a set of criteria from several aspects or dimensions that can be used to measure the performance of a business process. In general, there are four commonly used process performance measure including time, cost, quality, and flexibility [1].

In [10], the Metamodel Process Performance Indicators Notation and Tool (PPINOT) which can be used in defining Process Performance Indicators (PPI) is presented. According

to this study, the criteria that must be met in defining PPI are Specific, Measurable, Achievable, Relevant, and Time-bounded (SMART). Process Performance Indicators (PPI) is a performance measure that represents the success factors of a business process determined in a company or organization [11]. PPI can also be defined as quantifiable metrics that can be used to evaluate the efficiency and effectiveness of business processes.

The information that needs to be defined in the PPI is contained in several attributes in the PPINOT Metamodel, namely "Identifier: String", "Name: String", "relatedTo: Process", "Goals: String", "Definition: MeasureDefinition", "Target: Target", "Scope: Filter", "Responsible: Human Resource", "Informed: Human Resource", and "Comments: String" [10].

Then this research performs semantic mapping from Metamodel into Description Logistics to perform design-time analysis of the relationship between PPI and business process elements (BP elements) automatically. In addition, this study provides an example of defining PPI in a real case at the Andalusian Health Service Information Technology (IT) Department.

## III. METHODOLOGY

### A. Research Methodology

In general, there are several stages carried out in this research as shown in Fig. 1. This research begins by developing the PPM Framework. Second, we implement the PPM Framework by collecting the data and/or documents related to Surabaya COPACR based on the PPM Framework and interviews are carried out with the representatives of Surabaya COPACR. Third, we select business process based on several aspects of the criteria in Process Selection and Process Performance Measure. Fourth, we develop performance measurement indicator based on the business process architecture of Surabaya COPACR Service.

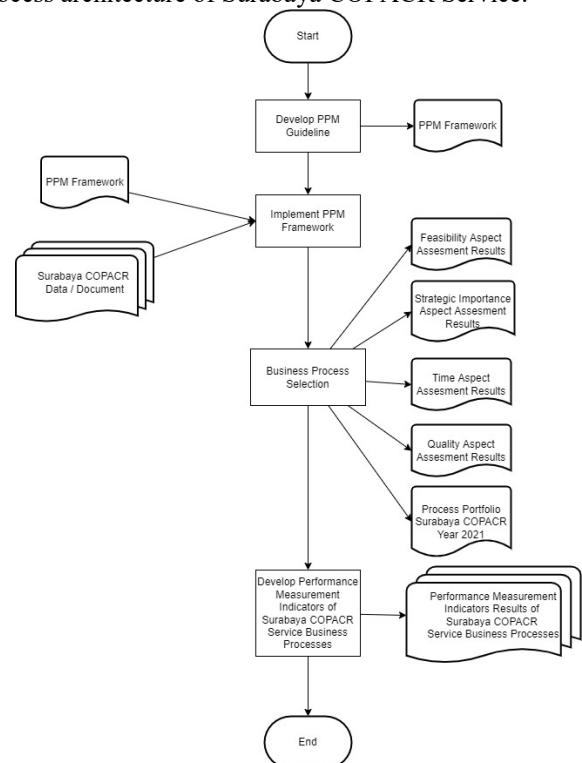


Fig. 1. Research Methodology Diagram

## B. Development of PPM Guideline

At this stage, development of PPM Framework is carried out by mapping from the PPINOT metamodel contained in the research article [10] against the data and documents from COPACR for each attribute in the PPINOT metamodel. Fig. 2 and Fig. 3 are the PPM guideline that has been developed in this research.<sup>1</sup>

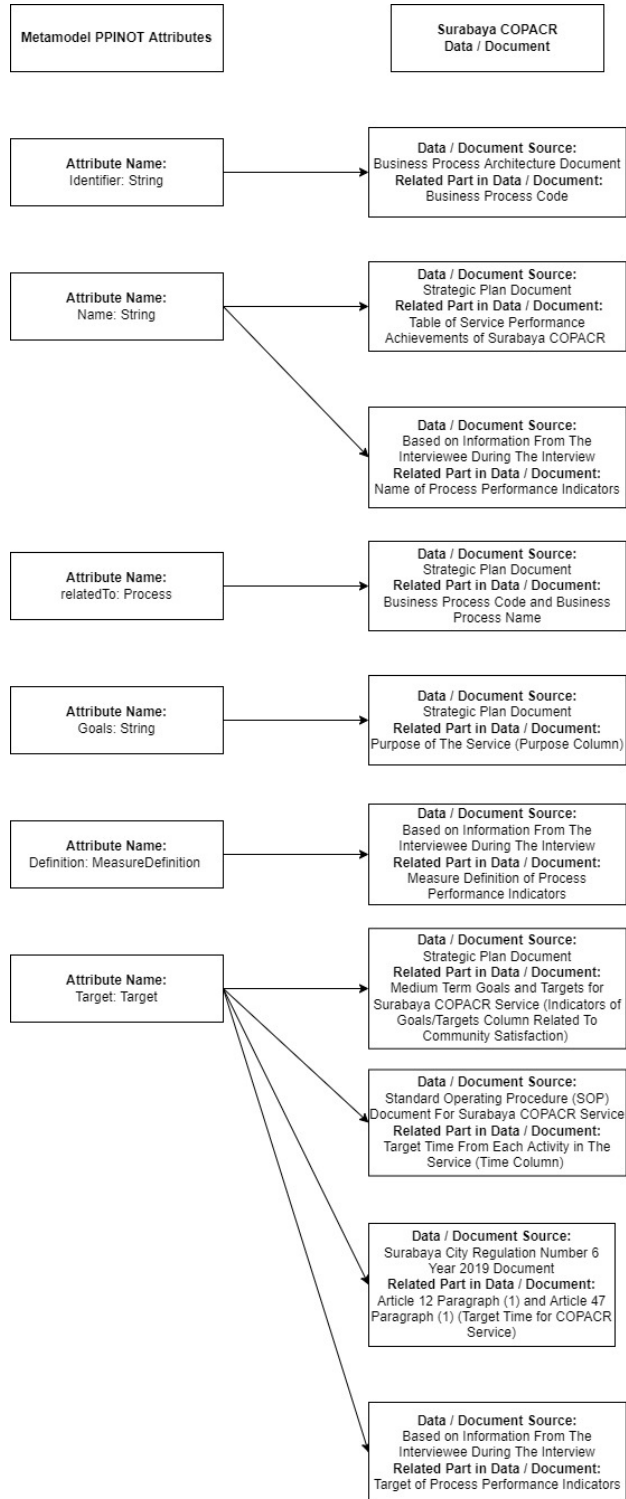


Fig. 2 The PPM Framework (Part 1)

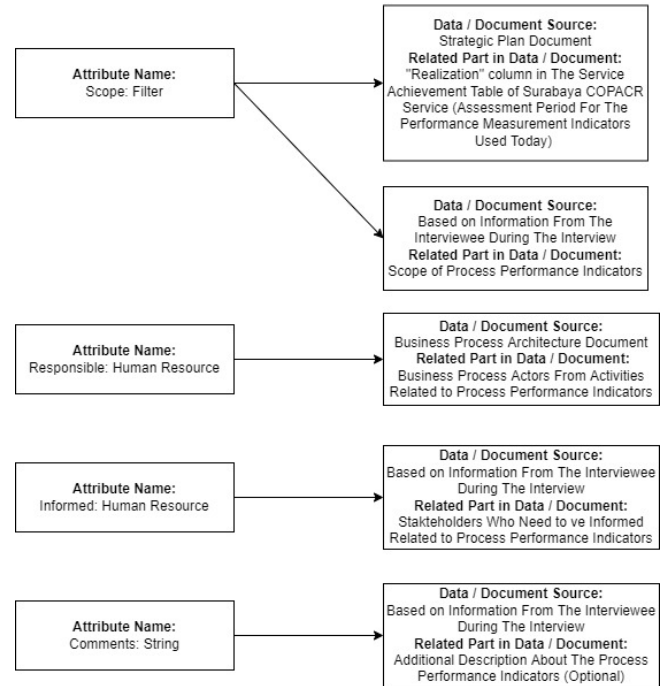


Fig. 3 The PPM Framework (Part 2)

First, the Surabaya COPACR business process was obtained from the Surabaya COPACR business process architecture document. This data is used as a reference in composing the performance measurement indicators of Surabaya COPACR service business process.

Data objectives of Surabaya COPACR service were obtained from the "Purpose" column contained in the strategic plan document from 2021 to 2026, namely "Improving the Quality of Public Services in the Field of Population Administration and Civil Registration". This data is used as a reference for information on the "Goals" column in the performance measurement indicators of Surabaya COPACR business process.

The performance measurement indicators used today by Surabaya COPACR are obtained from the column "Performance Indicators According to the Duties and Functions of Regional Apparatuses" in the Table of Service Performance Achievements of Surabaya COPACR contained in the strategic plan document from 2021 to 2026. This data is used as an initial reference for composing performance measurement indicators of Surabaya COPACR service business process.

The assessment period for the performance measurement indicators used today are obtained from the "Realization" column in the Service Achievement Table of Surabaya COPACR Service contained in the strategic plan document from 2021 to 2026. It is stated that the assessment period is every year. This data is used as a reference in determining the assessment period (scope) for composing the key performance measurement indicators (ID Code: KPPI) Surabaya COPACR service business process.

In general, there are 3 types of performance target data obtained in this study, namely the target data on community satisfaction for Surabaya COPACR service, target time per

<sup>1</sup> All Surabaya COPACR Data and Document in Fig. 2 and Fig. 3 are Directly Requested to Surabaya COPACR

activity, and target overall time from the beginning to the end of service delivery. The target data on community satisfaction value for Surabaya COPACR service is obtained from the column "Indicators of Goals/Targets" in the Table of Medium Term Goals and Targets for Surabaya COPACR Service contained in the strategic plan document from 2021 to 2026. The target value of community satisfaction for Surabaya COPACR service is 88%.

The target time per activity from each service is obtained from the Time column in the Standard Operating Procedure (SOP) document for Surabaya COPACR Service. This data is used as a reference in determining the target of business process performance measurement indicators (ID Code: PPI) related to "Time" for Surabaya COPACR service business process.

The target overall time from the beginning to the end of service delivery is obtained from Article 12 Paragraph (1) and Article 47 Paragraph (1) Surabaya City Regulation Number 6 Year 2019, which states that the target time for the COPACR service is 7 (seven) working days [12]. This target becomes a reference in determining the target of key business process performance measurement indicators (ID Code: KPPI) related to "Time" for Surabaya COPACR service business process. In addition, this data is also used as a target time for assessing the time aspect of Surabaya COPACR service business process.

In the business process selection stage, the following data are gathered:

1. Data on implementation time from the beginning to the end of the process (end to end process) in the Surabaya COPACR business process, was obtained from the timestamp data of Surabaya COPACR service application time in 2021. This data is used as one aspect (time aspect) in determining the health aspect (performance) of the business process.
2. Number of application defects in Surabaya COPACR service business process, obtained from the Surabaya COPACR service defect application data in 2021. This data is used as one aspect (quality aspect) in determining the health aspect (performance) of the business process.
3. The total application data for each business process of Surabaya COPACR service, obtained from the application data from the Surabaya COPACR service in 2021. This data is used to determine the value of the strategic importance aspect of the business process.
4. The average number of applications from the entire Surabaya COPACR service business process in 2021, obtained by dividing the total number of applications from all services in 2021 with the number of Surabaya COPACR services. From the calculation results, the average number of applications is 14525 applications. This data is used to determine the limit value of the strategic importance.

### C. Implementation of PPM Framework in COPACR

At this stage, the data and documents needed in this study were collected. Some of the data and documents needed are mentioned in the Section III Part B Interviews are carried out

with the representatives of Surabaya COPACR to determine the feasibility aspect of the business process and to validate the compiled performance measurement indicators. TABLE I provides detailed information from interviews.

TABLE I. Detailed Information of Data Collection

| Interview (date and duration) | Informant                                                                              | Information collected                                                                                                   |
|-------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>15 March 2022, 2 Hours</b> | Staff of General and Personnel Sub-Section (Unit Center of Excellence) Surabaya COPACR | Performance measurement indicators validation (1 <sup>st</sup> meeting), Information that required in the PPM Framework |
| <b>8 April 2022, 2 Hours</b>  | Staff of General and Personnel Sub-Section (Unit Center of Excellence) Surabaya COPACR | Performance measurement indicators validation (2 <sup>nd</sup> meeting), Information that required in the PPM Framework |
| <b>18 May 2022, 2 Hours</b>   | Staff of General and Personnel Sub-Section (Unit Center of Excellence) Surabaya COPACR | Performance measurement indicators validation (3 <sup>rd</sup> meeting), Information that required in the PPM Framework |
| <b>24 May 2022, 2 Hours</b>   | Staff of General and Personnel Sub-Section (Unit Center of Excellence) Surabaya COPACR | Performance measurement indicators validation (4 <sup>th</sup> meeting), Information that required in the PPM Framework |
| <b>2 June 2022, 1 Hour</b>    | Staff of General and Personnel Sub-Section (Unit Center of Excellence) Surabaya COPACR | Value of the feasibility aspect of the service business process                                                         |
| <b>13 June 2022, 2 Hours</b>  | Staff of General and Personnel Sub-Section (Unit Center of Excellence) Surabaya COPACR | Performance measurement indicators validation (5 <sup>th</sup> meeting), Information that required in the PPM Framework |

### D. Business Process Selection

The business process selection stage is selecting a business process based on several aspects of the criteria in Process Selection and Process Performance Measure. In general, there are 2 (two) core business processes that carried out by Surabaya COPACR, namely "Population Administration" and

"Civil Registration". In each of the core business process, there are several sub-service business processes that will be used and discussed in this research as shown in TABLE II. In addition, the KLAMPID application is an application that provides various types of services provided by the Surabaya COPACR.

TABLE II. Sub-Service Business Processes of Surabaya COPACR

| Core Business Process     | Sub-Service Business Process                                                                             |
|---------------------------|----------------------------------------------------------------------------------------------------------|
| Population Administration | Indonesian Citizens Transfer Certificate of Inter-City/Regency/Province                                  |
|                           | Indonesian Citizens Moving Out Certificates Between Cities / Districts / Provinces                       |
|                           | Indonesian Citizens Moving Out Certificates Between Cities / Districts / Provinces through Urban Village |
|                           | Biodata Change on Family Cards with the KLAMPID Application                                              |
|                           | Biodata Change on Family Cards with the KLAMPID Application through Urban Village                        |
| Civil Registration        | Birth Certificate                                                                                        |
|                           | Death Certificate                                                                                        |

In this stage there are several activities carried out. The first activity is making semi-structured questions as a reference in the interview. Then interviews were conducted with the staff of Surabaya COPACR to determine the feasibility aspect of the business process.

The next step is the assessment of the criteria aspects in the Process Selection and Process Performance Measure. Several aspects of the criteria assessed in the Process Selection are strategic importance, health, and feasibility of the business processes. The health aspect of Surabaya COPACR service business process can be measured based on performance data obtained from several aspects in the Process Performance Measure, namely the time aspect and the quality aspect. The cost aspect and the flexibility aspect are not measured in the health aspect of the service business process because the main focus of measuring the performance of Surabaya COPACR service business process is the speed (time aspect) and accuracy (quality aspect) of the application.

Then the Process Portfolio is developed based on the results of the assessment of several aspects of the criteria. From the results of the Process Portfolio, the focus area for business process improvement is determined. Finally, the business process selection is carried out by considering the results of the focus area and the feasibility aspect of the business process.

#### E. Performance Measurement Indicators of Surabaya COPACR Service Business Processes

In this stage the performance measurement indicators are developed based on the business process architecture of Surabaya COPACR Service. This stage begins by identifying the service performance measurement indicator currently used by Surabaya COPACR. The activity was continued by composing performance measurement indicators based on the

business process of Surabaya COPACR service. After that, detailed information from each performance measurement indicators are composed.

The results of composing performance measurement indicators and detailed information on each performance measurement indicators from service business processes are validated by the staff of Surabaya COPACR. If the performance measurement indicators and/or detailed information on each performance measurement indicators from service business processes is not valid, then correction and/or changes can be made based on the suggestions that had been given by the staff of Surabaya COPACR. If the proposed performance measurement indicators and detailed information on each performance measurement indicators from the service business process are valid, the results of performance measurement indicators and detailed information on each performance measurement indicators from service business process of Surabaya COPACR are obtained.

## IV. RESULTS AND DISCUSSION

### A. Business Process Selection Results

#### 1) Feasibility Aspect Assessment Results

From the interview activities that have been carried out, several results obtained from the feasibility aspect assessment of Surabaya COPACR service business process as shown in TABLE III.

TABLE III. Feasibility Score Results of Surabaya COPACR Service Business Process from Interview Activity

| Service Business Process                                                                                                              | Feasibility Score Results |
|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Birth Certificate Service Business Process                                                                                            | 5 - Very Feasible         |
| Death Certificate Service Business Process                                                                                            | 5 - Very Feasible         |
| Service Business Process for Indonesian Citizens Transfer Certificate of Inter-City/Regency/Province                                  | 4 - Feasible              |
| Service Business Process for Indonesian Citizens Moving Out Certificates Between Cities / Districts / Provinces                       | 5 - Very Feasible         |
| Service Business Process for Indonesian Citizens Moving Out Certificates Between Cities / Districts / Provinces through Urban Village | 4 - Feasible              |
| Service Business Process of Biodata Change on Family Cards with the KLAMPID Application                                               | 5 - Very Feasible         |
| Service Business Process of Biodata Change on Family Cards with the KLAMPID Application through Urban Village                         | 4 - Feasible              |

#### 2) Strategic Importance Aspect Assessment Results

The Birth Certificate Service Business Process is found to be the most strategically important according to the number 149869 application received.

#### 3) Time Aspect Assessment Results

The Business Process Services for Indonesian Citizens

Moving Out Certificates Between Cities / Districts / Provinces through Urban Village has the highest time aspect performance value (99.66%). This value (99.66%) means that this service business process has the highest percentage of applications completed on time according to the target (in this case the target time is 7 days). Meanwhile, business process with the lowest time aspect performance value is the Birth Certificate Service Business Process (87.37%). One of the reason is because it has a large total number of applications.

#### 4) Quality Aspect Assessment Results

From the quality aspect, the Birth Certificate Business Process has the highest quality aspect performance value (99.95%). This value (99.95%) means that this service business process has the highest percentage of non-defect applications. Meanwhile, service business process with the lowest quality aspect performance value is the Service Business Process of Biodata Change on Family Cards with the KLAMPID Application (89.76%). One of the reason is because it has a large total number of applications and there are many menu variations for submitting applications related to Family Card documents in the KLAMPID Application.

#### 5) Process Portfolio Surabaya COPACR

From the results of the assessment of time and health aspects, the value of business process is calculated by averaging the percentage of applications that match the target time limit (performance value from the time aspect) and the percentage non-defect application (performance value from quality aspect). The next activity is to build a Process Portfolio using the Tableau Desktop Professional Edition version 2022.1.2 software.

In the Process Portfolio, there are several boundaries that are determined to form the focus area for selecting business processes, namely the boundary of strategic importance aspects and boundary of health (performance) aspects. The boundary of strategic importance aspects determines whether the service business process is strategically important or not strategically important. The boundary value of strategic importance aspect used in Process Portfolio is based on the data average number of application for the entire Surabaya COPACR service business process in 2021, which is 14525 applications.

Then the boundary of health (performance) aspect will determine whether the performance of service business process is healthy or unhealthy. The boundary value of health (performance) aspect used in Process Portfolio is based on the median value of health aspect (performance) Surabaya COPACR service business process in 2021, which is 95.236%.

Based on the application of boundaries on aspects of strategic importance and health aspects, the focus areas for selecting service business processes are as shown in Fig. 4. The focus areas for selecting service business processes are service business processes that have unhealthy performance based on the value of health aspect (performance) and service business process that strategically important based on the value of strategic

importance aspect. The focus area for selecting the service business process was confirmed by the Head of Surabaya COPACR on June 17, 2022 that the focus area for selecting the service business process was valid and there were no suggestions regarding changes to the focus area for selecting the service business process.

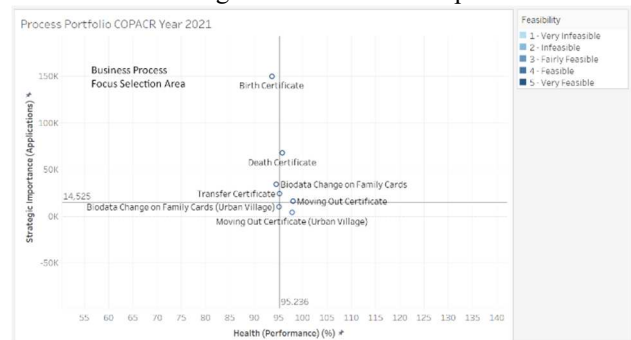


Fig. 4 Process Portfolio Surabaya COPACR Year 2021

Based on the results of the focus area for selecting service business process that has been obtained and taking into account the results of the feasibility aspect that in general the service business process is Very Feasible or Feasible to carry out business process improvement initiatives as in Fig. 4, the results obtained are recommendations for business process priorities that can be further analyzed and/or future improvement initiatives that can be carried out by the Surabaya COPACR, which are sequentially as follows (1) Birth Certificate, (2) Biodata Change on Family Cards with the KLAMPID Application, and (3) Indonesian Citizens Transfer Certificate of Inter-City/Regency/Province.

#### B. Performance Measurement Indicators for COPACR

One of the Surabaya COPACR service business processes that will be used as an example of the discussion in this research is Birth Certificate service business process. Based on the business process architecture of Birth Certificate Service, it can be seen that there are 16 activities and 7 decision points in the Birth Certificate service business process. Then from these activities and decision points, identification and composing of business process performance measurement indicators (ID Code: PPI) are carried out based on the time and quality aspects contained in the Process Performance Measure, which obtained as many as 25 business process performance measurement indicators that measure the performance of activities or decision points in the Birth Certificate service business process of Surabaya COPACR. Example of the business process performance measurement is provided in TABLE IV. In addition, there are 4 key business process performance measurement indicators that measure from the beginning to the end of the service business process (end to end process) (as shown in TABLE V).

TABLE IV. Business Process Performance Measurement Indicator (ID Code: PPI) Surabaya COPACR

| Name                                                                                    |
|-----------------------------------------------------------------------------------------|
| Speed of Response Time for Submitting a Verification of a Birth Certificate Application |
| Speed of Birth Certificate Application Verification Time                                |

|                                                                                   |
|-----------------------------------------------------------------------------------|
| Percentage of Verified Birth Certificate Applications with Duration > 1 Day       |
| Speed of Birth Certificate Application Validation Time                            |
| Percentage of Validated Birth Certificate Applications with Time Duration > 1 Day |
| Percentage of Invalid Birth Certificate Applications                              |

TABLE V. Key Business Process Performance Measurement Indicator (ID Code: KPPI) Surabaya COPACR

| Name                                                                                      |
|-------------------------------------------------------------------------------------------|
| Total Time Speed of Birth Certificate Service Process                                     |
| Percentage of Birth Certificate Applications with Total Processing Time Duration > 7 Days |
| Percentage of Defect Application for Birth Certificate                                    |
| Average Value of Community Satisfaction with Birth Certificate Services                   |

Based on the results of the identification of the current performance measurement indicators, it can be seen that the service performance measurement indicators currently used are performance measurement indicators that do not measure specifically per activity carried out in the service. Meanwhile, in this study, not only composing performance measurement indicators in general from the beginning to the end of the service business process (end to end process) (ID Code: KPPI) but also composing performance measurement indicators based on details per activity (activities) or decision points (ID code: PPI). Therefore, Surabaya COPACR can measure performance in more detail per activity or decision point carried out in the service business process.

This research has also linked the performance measurement indicators with the actors in the business process. In this case, this study links the performance measurement indicators with the actors who are responsible for the performance of these indicators (responsible) and the actors who need to be informed about the performance of these indicators (informed). Therefore, Surabaya COPACR can measure in more detail the performance of Surabaya COPACR's internal actors that involved in the service business process.

## V. CONCLUSION

In this study, we attempt to solve the challenges facing the Surabaya COPACR by applying BPM approach. Based on the interviews and assessment of criteria aspects in the Process Selection and Process Performance Measure of Surabaya COPACR business process, we found:

- Business processes that do not have the option to be submitted through the urban village have the highest feasibility to be changed.
- Birth Certificate Business Process is the most strategically important business process with the highest strategic importance aspect value.
- Business Process for Indonesian Citizens Moving Out Certificates Between Cities / Districts / Provinces through Urban Village has the highest performance related to time. Meanwhile, service business process with the slowest performance is the Birth Certificate Business Process.
- Birth Certificate Business Process has the highest quality performance. Meanwhile, business process with the lowest quality is the Biodata Change on Family Cards with the KLAMPID Application

Based on the results of building Process Portfolio of Surabaya COPACR business processes, priorities for business process improvement initiatives are as follows (1) Birth Certificate Service, (2) Biodata Change on Family Cards with the KLAMPID Application, and (3) Indonesian Citizens Transfer Certificate of Inter-City/Regency/Province processes.

From the results of composing performance measurement indicator, 29 performance measurement indicators of the Birth Certificate service business process were obtained, which consist of 25 business process performance measurement indicators (ID Code: PPI) that measure the performance of activities or decision points in the business process and 4 key business process performance measurement indicators (ID Code: KPPI) which measure performance in general from the beginning to the end of the service business process.

Based on these results, further research can be done to measure the impact of the process selection and process performance measures proposed in this study in Surabaya COPACR. Another avenue for further research is to conduct Process Selection, Process Performance Measure, and Process Portfolio in companies or organizations that focus on measuring performance on the cost and flexibility aspect.

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